

ForceDelta-VLA: Supplementary Material

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I. TRAINING AND ARCHITECTURE DETAILS

The temporal force-conditioned teacher is fully fine-tuned on eight NVIDIA A100 GPUs. All real-robot inference and runtime measurements use a single 48 GB NVIDIA RTX 4090D GPU. The teacher is trained for 50,000 steps with AdamW at a peak learning rate of 2×10^{-5} . It predicts $H=50$ actions at 30 Hz from the latest 100 ms of the 100 Hz wrench stream, and flow-matching inference uses $N_{\text{flow}}=10$ steps.

After freezing the teacher, we train the rank-32 null-mode adapter for 10,000 steps at a learning rate of 10^{-4} . The visual-language prefix is pooled into $M=16$ intent tokens. The residual policy comprises a causal force encoder and one pre-norm attention block with separate zero-initialized force and staleness output projections. It predicts $K=5$ corrections and is trained for 30,000 steps at a learning rate of 3×10^{-4} under replayed asynchronous schedules. The final implementation record will specify the intent-token pooling rule and the embedding used to mask the force token for the staleness head.

The action representation and rotational residual composition are defined in Sec. III-A of the main paper. At deployment, force-conditioned and staleness corrections are bounded independently in the normalized pose-action space before composition with the reference action. Let μ_A and $D_A = \text{diag}(\sigma_A)$ be the training-set action statistics. Absolute action coordinates use $\mathcal{N}(a) = D_A^{-1}(a - \mu_A)$, whereas action differences and the rebasing term use the scale-only map $\mathcal{N}_\Delta(\delta a) = D_A^{-1}\delta a$. The absolute-command map $\mathcal{R}$ de-normalizes the composed action before applying it relative to the reference query state. The final implementation record will verify these conventions and specify the per-head correction bounds.

Residual-policy training examples are anchored at the 30 Hz state/action timestamps while retaining every 100 Hz wrench sample in $\mathcal{F}_t$. Residual packets may be queried more frequently at deployment; the realized publication rate is measured directly rather than inferred from the waypoint rate.

A. Null-Mode Optimization

The learned null token and null-specific adapter are active only for null-mode queries. With the force-conditioned teacher parameters θ fixed, the null-mode parameters ϕ are optimized using the teacher's original flow-matching objective,

$$\mathcal{L}_{\text{null}} = \mathbb{E}_{\mathcal{D}, \eta, \epsilon} [\ell_{\text{FM}}(T_{\theta, \phi}^{\text{null}}, A_{t:t+H-1}^*; \eta, \epsilon)], \quad (1)$$

where η is flow time and ϵ is the initial flow noise. Full-force and null-force predictions used to construct a residual target share the same ϵ .

B. Residual-Loss Weighting

For a K -step residual chunk, the normalized temporal weights are

$$w_j = \frac{\beta^{j-1}}{\sum_{m=1}^K \beta^{m-1}}, \quad 0 < \beta \leq 1. \quad (2)$$

Thus, $\beta < 1$ emphasizes near-term corrections, $\beta = 1$ weights all steps equally, and $\sum_j w_j = 1$ keeps the loss scale stable as K changes. The loss is a mean-squared error in normalized pose-action coordinates.

II. ASYNCHRONOUS PACKET EXECUTION

A. Reference Packets

A reference query k , issued at t_k from state S_k , publishes the immutable packet $(A_{T,k,1:H}^{\text{null}}, Z_k^{\text{intent}}, S_k, t_k, \bar{t}_k)$ when its result becomes available at $\bar{t}_k$. Its samples carry intended execution times

$$t_{k,n}^{\text{ref}} = t_k + (n-1)/f_A, \quad n = 1, \dots, H. \quad (3)$$

At time t , the active reference is the freshest completed, non-superseded query,

$$k^*(t) = \arg \max_{k: \bar{t}_k \leq t} t_k. \quad (4)$$

An older query that completes later therefore cannot replace a newer active packet. Samples whose intended times precede $\bar{t}_{k^*}$ are skipped for direct execution; the cached chunk itself is not truncated. For $t \geq \bar{t}_{k^*}$, the reference pose $A_{k^*}^{\text{ref}}(t)$ is obtained by timestamp-based interpolation over the original timestamped chunk.

B. Residual Packets

A residual inference packet q , initiated at t_q^F , reads $k_q = k^*(t_q^F)$ together with $(\mathcal{F}_{t_q^F}, S_{t_q^F})$. It completes at $\bar{t}_q^F$ with force-conditioned and staleness correction chunks. Their samples are timestamped as

$$t_{q,j}^F = t_q^F + (j-1)/f_A, \quad j = 1, \dots, K. \quad (5)$$

Here f_A defines the waypoint spacing within a packet; residual query times t_q^F are scheduled independently. Step j is held over $[t_{q,j}^F, t_{q,j}^F + 1/f_A)$, and correction samples are not interpolated. At control time t , the executor selects the most recently initiated packet that has completed and remains valid, together with its active step:

$$q^*(t) = \arg \max_{q: \bar{t}_q^F \leq t < \bar{t}_q^F + K/f_A} t_q^F, \quad (6)$$

$$j(t; q^*) = 1 + \lfloor (t - t_{q^*}^F) f_A \rfloor.$$

TABLE I
SENSITIVITY TO THE RESIDUAL BOUND. RED ENTRIES ARE PLACEHOLDERS.

Bound	USB insertion		Drawer opening	
	Success (%)	Peak (N)	Success (%)	Peak (N)
0.5×	X	X	X	X
1×	X	X	X	X
2×	X	X	X	X

Any prefix that expires during inference is therefore skipped, while the remaining valid steps execute sequentially until a newer packet completes. Let $j^* = j(t; q^*)$. Within the selected interval, Eq. (1) of the main paper is evaluated using the residual packet’s query time and active step:

$$\begin{aligned} \mathcal{P}(A^{\text{cmd}}(t)) &\equiv \mathcal{P}(A_{t_{q^*,j^*}^F}^{\text{cmd}}) \\ &= \mathcal{P}\left(A_{k_{q^*}}^{\text{ref}}(t_{q^*,j^*}^F)\right) + \Delta\hat{A}_{q^*,j^*}^{\text{force}} + \Delta\hat{A}_{q^*,j^*}^{\text{stale}}. \end{aligned} \quad (7)$$

The two corrections are bounded independently and composed in normalized delta-action coordinates. The composed command remains referenced to $S_{k_{q^*}}$ and is converted to an absolute robot command using the mapping $\mathcal{R}$ defined in the main paper. A packet that completes after $t_q^F + K/f_A$ is rejected. If no valid residual packet remains, the executor invokes the platform’s safe hold or abort policy. **The final implementation record will specify gripper sampling and behavior beyond the last valid reference timestamp.**

III. ADDITIONAL EVALUATION PROTOCOLS

A. Metric Aggregation

Peak force is computed from every trial. For a bimanual trial, it is the maximum force norm over time and over both arms. Completion time is measured only on successful trials, from the first policy command to satisfaction of the task-specific success criterion. Platform-level values first aggregate trials within each task and then macro-average the four tasks. These conventions must be verified against the final evaluation code.

B. Seed-Wise Task Results

The final supplement will report the successful-trial count for each training seed, method, and task. Each count is out of ten trials and aggregates to the main paper’s success-rate table.

C. Residual-Bound Sensitivity

We evaluate the deployment correction bound at 0.5×, 1×, and 2× its default value on USB insertion and drawer opening. The force-conditioned and staleness heads are clipped independently, with per-arm clipping in the bimanual setting.